

Lovely Professional University, Punjab

Course Code	Course Title	Lectures	Tutorials	Practicals	Credits	
CSE121	ORIENTATION TO COMPUTING-II	2	0	0	2	
Course Weightage	ATT: 20 CA: 50 ETT: 30	Exam Category: X1: Mid Term Exam: Not Applicable – End Term Exam: All MCQ				
Course Focus	SKILL DEVELOPMENT,EMPLOYABILITY					

Course Outcomes :Through this course students should be able to

CO1 :: Understand the need for data science and big data along with its tools, job roles, and skill set

CO2 :: Discuss the use of AI and machine learning in different fields along with its tools, job roles, and skill set

CO3 :: Recognize the attacks, malware, tools, job roles, suspicious links, secure connection and skill set for cyber security

CO4 :: recognize various DevOps, Software testing tools with their applications, along with their job roles and skill sets.

CO5 :: discuss different types of cloud model implementation, services along with their tools, job roles, and skill set.

CO6 :: compare usefulness of various front end, backend technologies along with their tools, job roles, and skill set.

	Reference Books (R)		
Sr No	Title	Author	Publisher Name
R-1	BIG DATA	ANIL MAHESHWARI	MC GRAW HILL
R-2	PRINCIPLES OF SOFT COMPUTING	S.N. SIVANANDAM, S.N. DEEPA	WILEY
R-3	HTML, CSS, AND JAVASCRIPT ALL IN ONE, SAMS TEACH YOURSELF	JULIE C. MELONI, JENNIFER KYRNIN	PEARSON
R-4	CLOUD COMPUTING: FUNDAMENTALS, INDUSTRY APPROACH AND TRENDS	RISHABH SHARMA	WILEY

Relevant Websites (RW)		
Sr No	(Web address) (only if relevant to the course)	Salient Features
RW-1	https://www.oracle.com/big-data/	Big Data
RW-2	https://www.ibm.com/in-en/analytics/big-data-analytics	Big Data Analytics
RW-3	https://www.ibm.com/topics/data-science	Data Science
RW-4	https://aws.amazon.com/what-is/data-science/	Data science future

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RW-5	https://www.cisco.com/c/en_in/products/security/what-is-cybersecurity.html	Cyber Security
RW-6	https://www.ibm.com/topics/cloud-computing	Cloud Computing
RW-7	https://www.salesforce.com/in/learning-centre/tech/cloudcomputing/	Salesforce Cloud
RW-8	https://cloud.google.com/learn/what-is-cloud-computing	Google cloud def
RW-9	https://aws.amazon.com/devops/what-is-devops/	DevOps
RW-10	https://www.mongodb.com/languages/full-stack-development	Full Stack Development
RW-11	https://www.ibm.com/topics/machine-learning	Machine Learning
RW-12	https://owasp.org/www-community/attacks/Brute_force_attack	Brute force attack

LTP week distribution: (LTP Weeks)	
Weeks before MTE	7
Weeks After MTE	7
Spill Over (Lecture)	4

Detailed Plan For Lectures

Week Number	Lecture Number	Broad Topic(Sub Topic)	Chapters/Sections of Text/reference books	Other Readings, Relevant Websites, Audio Visual Aids, software and Virtual Labs	Lecture Description	Learning Outcomes	Pedagogical Tool Demonstration/ Case Study / Images / animation / ppt etc. Planned	Live Examples
Week 1	Lecture 1	Data Science & Big Data (Data science and its need)	R-1	RW-1 RW-2 RW-3 RW-4	L1: Lecture zero to be delivered so that student will understand why they need to study this subject. L2: Discuss about the Introduction, tools, applications, job roles, and skill set of data science and Big data	Students would be able to understand the tools, applications, job roles, and skill set of data science and Big data	Demonstration with PowerPoint presentation	Business, health, tracking customer spending habit, shopping behavior, recommendations, Education sector

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

Week 1	Lecture 1	Data Science & Big Data (Applications of data science/Big data)	R-1	RW-1 RW-2 RW-3 RW-4	L1: Lecture zero to be delivered so that student will understand why they need to study this subject. L2: Discuss about the Introduction, tools, applications, job roles, and skill set of data science and Big data	Students would be able to understand the tools, applications, job roles, and skill set of data science and Big data	Demonstration with PowerPoint presentation	Business, health, tracking customer spending habit, shopping behavior, recommendations, Education sector
	Lecture 2	Data Science & Big Data (Data science Lifecycle with use case)	R-1	RW-1 RW-2 RW-3 RW-4	Discuss about the Data science Lifecycle with use case, 3 V's of Big data, Challenges and Skill needed for Big data	Students would be able to understand the Data science Lifecycle with use case, 3 V's of Big data, Challenges and Skill needed for Big data	Demonstration with PowerPoint presentation	Business, health, tracking customer spending habit, shopping behavior, recommendations, Education sector
		Data Science & Big Data (Big data and its 3Vs)	R-1	RW-1 RW-2 RW-3 RW-4	Discuss about the Data science Lifecycle with use case, 3 V's of Big data, Challenges and Skill needed for Big data	Students would be able to understand the Data science Lifecycle with use case, 3 V's of Big data, Challenges and Skill needed for Big data	Demonstration with PowerPoint presentation	Business, health, tracking customer spending habit, shopping behavior, recommendations, Education sector
		Data Science & Big Data (Challenges of Big data)	R-1	RW-1 RW-2 RW-3 RW-4	Discuss about the Data science Lifecycle with use case, 3 V's of Big data, Challenges and Skill needed for Big data	Students would be able to understand the Data science Lifecycle with use case, 3 V's of Big data, Challenges and Skill needed for Big data	Demonstration with PowerPoint presentation	Business, health, tracking customer spending habit, shopping behavior, recommendations, Education sector
		Data Science & Big Data (Skill needed for Big data)	R-1	RW-1 RW-2 RW-3 RW-4	Discuss about the Data science Lifecycle with use case, 3 V's of Big data, Challenges and Skill needed for Big data	Students would be able to understand the Data science Lifecycle with use case, 3 V's of Big data, Challenges and Skill needed for Big data	Demonstration with PowerPoint presentation	Business, health, tracking customer spending habit, shopping behavior, recommendations, Education sector

Week 2	Lecture 3	Data Science & Big Data (Tools usage like Apache Hadoop, Tableau, R language, Excel)	R-1	RW-1 RW-2 RW-3 RW-4	Discuss about the Introduction of tools, Big data on cloud, Job roles, and skill set of data science and Big data	Students would be able to understand the tools, Big data on cloud, Job roles, and skill set of data science and Big data	Demonstration with PowerPoint presentation	Business, health, tracking customer spending habit, shopping behavior, recommendation s, Education sector
		Data Science & Big Data (Big Data on the Cloud)	R-1	RW-1 RW-2 RW-3 RW-4	Discuss about the Introduction of tools, Big data on cloud, Job roles, and skill set of data science and Big data	Students would be able to understand the tools, Big data on cloud, Job roles, and skill set of data science and Big data	Demonstration with PowerPoint presentation	Business, health, tracking customer spending habit, shopping behavior, recommendation s, Education sector
		Data Science & Big Data (Use of Big Data in different areas)	R-1	RW-1 RW-2 RW-3 RW-4	Discuss about the Introduction of tools, Big data on cloud, Job roles, and skill set of data science and Big data	Students would be able to understand the tools, Big data on cloud, Job roles, and skill set of data science and Big data	Demonstration with PowerPoint presentation	Business, health, tracking customer spending habit, shopping behavior, recommendation s, Education sector
		Data Science & Big Data (Job roles and skillset for Data science and Big data)	R-1	RW-1 RW-2 RW-3 RW-4	Discuss about the Introduction of tools, Big data on cloud, Job roles, and skill set of data science and Big data	Students would be able to understand the tools, Big data on cloud, Job roles, and skill set of data science and Big data	Demonstration with PowerPoint presentation	Business, health, tracking customer spending habit, shopping behavior, recommendation s, Education sector
	Lecture 4	Artificial Intelligence & Machine Learning (Introduction to AI)	R-2	RW-11	Discuss about the Introduction, tools, applications, job roles, and skill set of AI and machine learning	Students would be able to understand the tools, applications, job roles, and skill set of AI and machine learning	Demonstration with PowerPoint presentation	Smart home, agriculture, Speech recognition, Medical diagnosis, Statistical arbitrage, Predictive analytics

Week 3	Lecture 5	Artificial Intelligence & Machine Learning(ML and Deep Learning)	R-2	RW-11	Discuss about the Introduction, tools, applications, job roles, and skill set of AI and machine learning	Students would be able to understand the tools, applications, job roles, and skill set of AI and machine learning	Demonstration with PowerPoint presentation	Smart home, agriculture, Speech recognition, Medical diagnosis, Statistical arbitrage, Predictive analytics
		Artificial Intelligence & Machine Learning(Expert systems)	R-2	RW-11	Discuss about the Introduction, tools, applications, job roles, and skill set of AI and machine learning	Students would be able to understand the tools, applications, job roles, and skill set of AI and machine learning	Demonstration with PowerPoint presentation	Smart home, agriculture, Speech recognition, Medical diagnosis, Statistical arbitrage, Predictive analytics
		Artificial Intelligence & Machine Learning(Fuzzy systems)	R-2	RW-11	Discuss about the Introduction, tools, applications, job roles, and skill set of AI and machine learning	Students would be able to understand the tools, applications, job roles, and skill set of AI and machine learning	Demonstration with PowerPoint presentation	Smart home, agriculture, Speech recognition, Medical diagnosis, Statistical arbitrage, Predictive analytics
	Lecture 6	Artificial Intelligence & Machine Learning (Augmented Reality)	R-2	RW-11	Discuss about the Introduction, tools, applications, job roles, and skill set of AI and machine learning	Students would be able to understand the tools, applications, job roles, and skill set of AI and machine learning	Demonstration with PowerPoint presentation	Smart home, agriculture, Speech recognition, Medical diagnosis, Statistical arbitrage, Predictive analytics

Week 3	Lecture 6	Artificial Intelligence & Machine Learning(Use of AI in different fields - NLP, Healthcare, Agriculture, Social media monitoring)	R-2	RW-11	Discuss about the Introduction, tools, applications, job roles, and skill set of AI and machine learning	Students would be able to understand the tools, applications, job roles, and skill set of AI and machine learning	Demonstration with PowerPoint presentation	Smart home, agriculture, Speech recognition, Medical diagnosis, Statistical arbitrage, Predictive analytics
Week 4	Lecture 7	Artificial Intelligence & Machine Learning(Tools and techniques for implementing AI)	R-2	RW-11	Discuss about the Introduction, tools, applications, job roles, and skill set of AI and machine learning	Students would be able to understand the tools, applications, job roles, and skill set of AI and machine learning	Demonstration with PowerPoint presentation	Smart home, agriculture, Speech recognition, Medical diagnosis, Statistical arbitrage, Predictive analytics
		Artificial Intelligence & Machine Learning(Google Translator)	R-2	RW-11	Discuss about the Introduction, tools, applications, job roles, and skill set of AI and machine learning	Students would be able to understand the tools, applications, job roles, and skill set of AI and machine learning	Demonstration with PowerPoint presentation	Smart home, agriculture, Speech recognition, Medical diagnosis, Statistical arbitrage, Predictive analytics
		Artificial Intelligence & Machine Learning (Driverless Car)	R-2	RW-11	Discuss about the Introduction, tools, applications, job roles, and skill set of AI and machine learning	Students would be able to understand the tools, applications, job roles, and skill set of AI and machine learning	Demonstration with PowerPoint presentation	Smart home, agriculture, Speech recognition, Medical diagnosis, Statistical arbitrage, Predictive analytics

Week 4	Lecture 7	Artificial Intelligence & Machine Learning(ALEXA)	R-2	RW-11	Discuss about the Introduction, tools, applications, job roles, and skill set of AI and machine learning	Students would be able to understand the tools, applications, job roles, and skill set of AI and machine learning	Demonstration with PowerPoint presentation	Smart home, agriculture, Speech recognition, Medical diagnosis, Statistical arbitrage, Predictive analytics
		Artificial Intelligence & Machine Learning(Siri)	R-2	RW-11	Discuss about the Introduction, tools, applications, job roles, and skill set of AI and machine learning	Students would be able to understand the tools, applications, job roles, and skill set of AI and machine learning	Demonstration with PowerPoint presentation	Smart home, agriculture, Speech recognition, Medical diagnosis, Statistical arbitrage, Predictive analytics
		Artificial Intelligence & Machine Learning (ChatGPT)	R-2	RW-11	Discuss about the Introduction, tools, applications, job roles, and skill set of AI and machine learning	Students would be able to understand the tools, applications, job roles, and skill set of AI and machine learning	Demonstration with PowerPoint presentation	Smart home, agriculture, Speech recognition, Medical diagnosis, Statistical arbitrage, Predictive analytics
	Lecture 8	Artificial Intelligence & Machine Learning(Current trends and opportunities)	R-2	RW-11	Discuss about the Introduction, tools, applications, job roles, and skill set of AI and machine learning	Students would be able to understand the tools, applications, job roles, and skill set of AI and machine learning	Demonstration with PowerPoint presentation	Smart home, agriculture, Speech recognition, Medical diagnosis, Statistical arbitrage, Predictive analytics

Week 4	Lecture 8	Artificial Intelligence & Machine Learning(Job roles and skillset for AI and ML)	R-2	RW-11	Discuss about the Introduction, tools, applications, job roles, and skill set of AI and machine learning	Students would be able to understand the tools, applications, job roles, and skill set of AI and machine learning	Demonstration with PowerPoint presentation	Smart home, agriculture, Speech recognition, Medical diagnosis, Statistical arbitrage, Predictive analytics
Week 5	Lecture 9				Certification - MOOCs 1			
	Lecture 10	Introduction to Cyber Security(Introduction to cyber security)		RW-5	Discuss about the Types of malware and Types of attacks	Students would be able to understand the different Types of malware and Types of attacks.	Demonstration with PowerPoint presentation	Malware, Ransomware Attacks, Man in the Middle Attacks, Distributed Denial of Service (DDoS), Password Attacks, Drive-By Download Attacks
		Introduction to Cyber Security(Information security concepts)		RW-5	Discuss about the Types of malware and Types of attacks	Students would be able to understand the different Types of malware and Types of attacks.	Demonstration with PowerPoint presentation	Malware, Ransomware Attacks, Man in the Middle Attacks, Distributed Denial of Service (DDoS), Password Attacks, Drive-By Download Attacks
		Introduction to Cyber Security(Threats)		RW-5	Discuss about the Types of malware and Types of attacks	Students would be able to understand the different Types of malware and Types of attacks.	Demonstration with PowerPoint presentation	Malware, Ransomware Attacks, Man in the Middle Attacks, Distributed Denial of Service (DDoS), Password Attacks, Drive-By Download Attacks

Week 5	Lecture 10	Introduction to Cyber Security(Types of attacks)		RW-5	Discuss about the Types of malware and Types of attacks	Students would be able to understand the different Types of malware and Types of attacks.	Demonstration with PowerPoint presentation	Malware, Ransomware Attacks, Man in the Middle Attacks, Distributed Denial of Service (DDoS), Password Attacks, Drive-By Download Attacks
Week 6	Lecture 11	Introduction to Cyber Security(Use of cyber security in different industries like Healthcare, Manufacturing, E-commerce)		RW-5	Discuss about the Use of cyber security in different industries like Healthcare, Manufacturing, E-commerce. Also discuss about various Tools for cyber security assessment like nmap, wireshark, maltego	Students would be able to understand the Use of cyber security in different industries like Healthcare, Manufacturing, E-commerce. Also understand the various Tools for cyber security assessment like nmap, wireshark, maltego	Demonstration with PowerPoint presentation	Malware, Ransomware Attacks, Man in the Middle Attacks, Distributed Denial of Service (DDoS), Password Attacks, Drive-By Download Attacks
		Introduction to Cyber Security(Tools for cyber security assessment like nmap, wireshark, maltego)		RW-5	Discuss about the Use of cyber security in different industries like Healthcare, Manufacturing, E-commerce. Also discuss about various Tools for cyber security assessment like nmap, wireshark, maltego	Students would be able to understand the Use of cyber security in different industries like Healthcare, Manufacturing, E-commerce. Also understand the various Tools for cyber security assessment like nmap, wireshark, maltego	Demonstration with PowerPoint presentation	Malware, Ransomware Attacks, Man in the Middle Attacks, Distributed Denial of Service (DDoS), Password Attacks, Drive-By Download Attacks
	Lecture 12	Introduction to Cyber Security(AI based cyber threat intelligence solutions)		RW-5	Discuss about the Introduction, tools, applications, job roles, and skill set of cyber security	Students would be able to understand the tools, applications, job roles, and skill set of cyber security	Demonstration with PowerPoint presentation	Malware, Ransomware Attacks, Man in the Middle Attacks, Distributed Denial of Service (DDoS), Password Attacks, Drive-By Download Attacks

Week 6	Lecture 12	Introduction to Cyber Security(Cyber security opportunities in market and skillset)		RW-5	Discuss about the Introduction, tools, applications, job roles, and skill set of cyber security	Students would be able to understand the tools, applications, job roles, and skill set of cyber security	Demonstration with PowerPoint presentation	Malware, Ransomware Attacks, Man in the Middle Attacks, Distributed Denial of Service (DDoS), Password Attacks, Drive-By Download Attacks
Week 7	Lecture 13	Secure web-browsing (Recognize a secure connection)		RW-12	Discuss about the process of recognizing suspicious links and untrusted source warnings.	Students will learn about the process of recognizing suspicious links and recognizing untrusted source warnings.	Demonstration with powerpoint presentation	
		Secure web-browsing (Recognize suspicious links,)		RW-12	Discuss about the process of recognizing suspicious links and untrusted source warnings.	Students will learn about the process of recognizing suspicious links and recognizing untrusted source warnings.	Demonstration with powerpoint presentation	
		Secure web-browsing (Update Browsers and plugins)		RW-12	Discuss about the process of recognizing suspicious links and untrusted source warnings.	Students will learn about the process of recognizing suspicious links and recognizing untrusted source warnings.	Demonstration with powerpoint presentation	
		Secure web-browsing (Recognize untrusted source warnings)		RW-12	Discuss about the process of recognizing suspicious links and untrusted source warnings.	Students will learn about the process of recognizing suspicious links and recognizing untrusted source warnings.	Demonstration with powerpoint presentation	
		Secure web-browsing(social media security (facebook, whatsapp, email).)		RW-12	Discuss about the process of recognizing suspicious links and untrusted source warnings.	Students will learn about the process of recognizing suspicious links and recognizing untrusted source warnings.	Demonstration with powerpoint presentation	
		SPILL OVER						
Week 7	Lecture 14				Spill Over			
		MID-TERM						

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Week 8	Lecture 15	DevOps & Software Engineering(Introduction to DevOps)		RW-9	Discuss about the Introduction, tools, applications, job roles, and skill set of DevOps and Software Engineering	Students would be able to understand the tools, applications, job roles, and skill set of DevOps and Software Engineering	Demonstration with PowerPoint presentation	Microservices, continuous delivery, Network cycling
		DevOps & Software Engineering(DevOps Vs Traditional Software Development Models)		RW-9	Discuss about the Introduction, tools, applications, job roles, and skill set of DevOps and Software Engineering	Students would be able to understand the tools, applications, job roles, and skill set of DevOps and Software Engineering	Demonstration with PowerPoint presentation	Microservices, continuous delivery, Network cycling
		DevOps & Software Engineering(DevOps Tools : Git, Docker, Selenium, Mavin, Puppet, Ansible, Kubernetes, Nagios)		RW-9	Discuss about the Introduction, tools, applications, job roles, and skill set of DevOps and Software Engineering	Students would be able to understand the tools, applications, job roles, and skill set of DevOps and Software Engineering	Demonstration with PowerPoint presentation	Microservices, continuous delivery, Network cycling
	Lecture 16	DevOps & Software Engineering(Fundamentals of testing)		RW-9	Discuss about the Introduction, tools, applications, job roles, and skill set of DevOps and Software Engineering	Students would be able to understand the tools, applications, job roles, and skill set of DevOps and Software Engineering	Demonstration with PowerPoint presentation	Microservices, continuous delivery, Network cycling
		DevOps & Software Engineering(Objectives of Testing)		RW-9	Discuss about the Introduction, tools, applications, job roles, and skill set of DevOps and Software Engineering	Students would be able to understand the tools, applications, job roles, and skill set of DevOps and Software Engineering	Demonstration with PowerPoint presentation	Microservices, continuous delivery, Network cycling
		DevOps & Software Engineering(Types of Testing)		RW-9	Discuss about the Introduction, tools, applications, job roles, and skill set of DevOps and Software Engineering	Students would be able to understand the tools, applications, job roles, and skill set of DevOps and Software Engineering	Demonstration with PowerPoint presentation	Microservices, continuous delivery, Network cycling
Week 9	Lecture 17	DevOps & Software Engineering(Levels of testing)		RW-9	Discuss about the Introduction, tools, applications, job roles, and skill set of DevOps and Software Engineering	Students would be able to understand the tools, applications, job roles, and skill set of DevOps and Software Engineering	Demonstration with PowerPoint presentation	Microservices, continuous delivery, Network cycling
		DevOps & Software Engineering(Applications of software testing in IT companies)		RW-9	Discuss about the Introduction, tools, applications, job roles, and skill set of DevOps and Software Engineering	Students would be able to understand the tools, applications, job roles, and skill set of DevOps and Software Engineering	Demonstration with PowerPoint presentation	Microservices, continuous delivery, Network cycling

Week 9	Lecture 17	DevOps & Software Engineering(Career opportunities in the field of DevOps and software testing with skillset)		RW-9	Discuss about the Introduction, tools, applications, job roles, and skill set of DevOps and Software Engineering	Students would be able to understand the tools, applications, job roles, and skill set of DevOps and Software Engineering	Demonstration with PowerPoint presentation	Microservices, continuous delivery, Network cycling
	Lecture 18				Certification - MOOCs 2			
Week 10	Lecture 19	Introduction to Cloud Computing(Introduction to cloud computing)	R-4	RW-6 RW-7 RW-8	Discuss about the Introduction, tools, applications, job roles, and skill set of Cloud computing	Students would be able to understand the tools, applications, job roles, and skill set of Cloud computing	Demonstration with PowerPoint presentation	Software-as-a-Service (Saas): Salesforce Infrastructureas-a-Service (Saas): DigitalOcean Platform-as-a-Service (Saas): AWS File Sharing + Data Storage: Dropbox Big Data Analysis: Civis Analytics Data Governance: Carbonite Cybersecurity: Forcepoint
		Introduction to Cloud Computing(Uses of cloud computing in applications services)	R-4	RW-6 RW-7 RW-8	Discuss about the Introduction, tools, applications, job roles, and skill set of Cloud computing	Students would be able to understand the tools, applications, job roles, and skill set of Cloud computing	Demonstration with PowerPoint presentation	Software-as-a-Service (Saas): Salesforce Infrastructureas-a-Service (Saas): DigitalOcean Platform-as-a-Service (Saas): AWS File Sharing + Data Storage: Dropbox Big Data Analysis: Civis Analytics Data Governance: Carbonite Cybersecurity: Forcepoint

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Week 10	Lecture 19	Introduction to Cloud Computing(Platform deployments)	R-4	RW-6 RW-7 RW-8	Discuss about the Introduction, tools, applications, job roles, and skill set of Cloud computing	Students would be able to understand the tools, applications, job roles, and skill set of Cloud computing	Demonstration with PowerPoint presentation	Software-as-a-Service (Saas): Salesforce Infrastructureas-a-Service (Saas): DigitalOcean Platform-as-a-Service (Saas): AWS File Sharing + Data Storage: Dropbox Big Data Analysis: Civis Analytics Data Governance: Carbonite Cybersecurity: Forcepoint
	Lecture 20	Introduction to Cloud Computing(Types of cloud model implementations)	R-4	RW-6 RW-7 RW-8	Discuss about the Introduction, tools, applications, job roles, and skill set of Cloud computing	Students would be able to understand the tools, applications, job roles, and skill set of Cloud computing	Demonstration with PowerPoint presentation	Software-as-a-Service (Saas): Salesforce Infrastructureas-a-Service (Saas): DigitalOcean Platform-as-a-Service (Saas): AWS File Sharing + Data Storage: Dropbox Big Data Analysis: Civis Analytics Data Governance: Carbonite Cybersecurity: Forcepoint

Week 10	Lecture 20	Introduction to Cloud Computing(Types of cloud services)	R-4	RW-6 RW-7 RW-8	Discuss about the Introduction, tools, applications, job roles, and skill set of Cloud computing	Students would be able to understand the tools, applications, job roles, and skill set of Cloud computing	Demonstration with PowerPoint presentation	Software-as-a-Service (Saas): Salesforce Infrastructureas-a-Service (Saas): DigitalOcean Platform-as-a-Service (Saas): AWS File Sharing + Data Storage: Dropbox Big Data Analysis: Civis Analytics Data Governance: Carbonite Cybersecurity: Forcepoint
Week 11	Lecture 21	Introduction to Cloud Computing(Data analytics)	R-4	RW-6 RW-7 RW-8	Discuss about the Introduction, tools, applications, job roles, and skill set of Cloud computing	Students would be able to understand the tools, applications, job roles, and skill set of Cloud computing	Demonstration with PowerPoint presentation	
	Lecture 22	Introduction to Cloud Computing(Virtualization)	R-4	RW-6 RW-7 RW-8	Discuss about the Introduction, tools, applications, job roles, and skill set of Cloud computing	Students would be able to understand the tools, applications, job roles, and skill set of Cloud computing	Demonstration with PowerPoint presentation	Software-as-a-Service (Saas): Salesforce Infrastructureas-a-Service (Saas): DigitalOcean Platform-as-a-Service (Saas): AWS File Sharing + Data Storage: Dropbox Big Data Analysis: Civis Analytics Data Governance: Carbonite Cybersecurity: Forcepoint

Week 11	Lecture 22	Introduction to Cloud Computing(Tools and techniques for implementing)	R-4	RW-6 RW-7 RW-8	Discuss about the Introduction, tools, applications, job roles, and skill set of Cloud computing	Students would be able to understand the tools, applications, job roles, and skill set of Cloud computing	Demonstration with PowerPoint presentation	Software-as-a-Service (Saas): Salesforce Infrastructureas-a-Service (Saas): DigitalOcean Platform-as-a-Service (Saas): AWS File Sharing + Data Storage: Dropbox Big Data Analysis: Civis Analytics Data Governance: Carbonite Cybersecurity: Forcepoint
		Introduction to Cloud Computing(Job roles and skillset for cloud computing)	R-4	RW-6 RW-7 RW-8	Discuss about the Introduction, tools, applications, job roles, and skill set of Cloud computing	Students would be able to understand the tools, applications, job roles, and skill set of Cloud computing	Demonstration with PowerPoint presentation	Software-as-a-Service (Saas): Salesforce Infrastructureas-a-Service (Saas): DigitalOcean Platform-as-a-Service (Saas): AWS File Sharing + Data Storage: Dropbox Big Data Analysis: Civis Analytics Data Governance: Carbonite Cybersecurity: Forcepoint
Week 12	Lecture 23	Introduction to Full Stack Web Development and UI/UX(Introduction to Web Development)	R-3	RW-10	Discuss about the Introduction, tools, applications, job roles, and skill set of Full stack web development	Students would be able to understand the tools, applications, job roles, and skill set of Full stack web development	Demonstration with PowerPoint presentation	website development

Week 12	Lecture 23	Introduction to Full Stack Web Development and UI/UX(User Interface Design)	R-3	RW-10	Discuss about the Introduction, tools, applications, job roles, and skill set of Full stack web development	Students would be able to understand the tools, applications, job roles, and skill set of Full stack web development	Demonstration with PowerPoint presentation	website development
		Introduction to Full Stack Web Development and UI/UX(frontend)	R-3	RW-10	Discuss about the Introduction, tools, applications, job roles, and skill set of Full stack web development	Students would be able to understand the tools, applications, job roles, and skill set of Full stack web development	Demonstration with PowerPoint presentation	website development
	Lecture 24				Portfolio			
Week 13	Lecture 25	Introduction to Full Stack Web Development and UI/UX(backend)	R-3	RW-10	Discuss about the Introduction, tools, applications, job roles, and skill set of Full stack web development	Students would be able to understand the tools, applications, job roles, and skill set of Full stack web development	Demonstration with PowerPoint presentation	website development
		Introduction to Full Stack Web Development and UI/UX(databases)	R-3	RW-10	Discuss about the Introduction, tools, applications, job roles, and skill set of Full stack web development	Students would be able to understand the tools, applications, job roles, and skill set of Full stack web development	Demonstration with PowerPoint presentation	website development
		Introduction to Full Stack Web Development and UI/UX(CRUD applications)	R-3	RW-10	Discuss about the Introduction, tools, applications, job roles, and skill set of Full stack web development	Students would be able to understand the tools, applications, job roles, and skill set of Full stack web development	Demonstration with PowerPoint presentation	website development
	Lecture 26	Introduction to Full Stack Web Development and UI/UX(Languages such as HTML, CSS, PHP, Java Scripts, and frameworks)	R-3	RW-10	Discuss about the Introduction, tools, applications, job roles, and skill set of Full stack web development	Students would be able to understand the tools, applications, job roles, and skill set of Full stack web development	Demonstration with PowerPoint presentation	website development
		Introduction to Full Stack Web Development and UI/UX(by using VS code tool)	R-3	RW-10	Discuss about the Introduction, tools, applications, job roles, and skill set of Full stack web development	Students would be able to understand the tools, applications, job roles, and skill set of Full stack web development	Demonstration with PowerPoint presentation	website development

Week 13	Lecture 26	Introduction to Full Stack Web Development and UI/UX(Single page applications (SPA))	R-3	RW-10	Discuss about the Introduction, tools, applications, job roles, and skill set of Full stack web development	Students would be able to understand the tools, applications, job roles, and skill set of Full stack web development	Demonstration with PowerPoint presentation	website development
Week 14	Lecture 27	Introduction to Full Stack Web Development and UI/UX(Responsive web design)	R-3	RW-10	Discuss about the Introduction, tools, applications, job roles, and skill set of Full stack web development	Students would be able to understand the tools, applications, job roles, and skill set of Full stack web development	Demonstration with PowerPoint presentation	website development
		Introduction to Full Stack Web Development and UI/UX(mobile-first development)	R-3	RW-10	Discuss about the Introduction, tools, applications, job roles, and skill set of Full stack web development	Students would be able to understand the tools, applications, job roles, and skill set of Full stack web development	Demonstration with PowerPoint presentation	website development
		Introduction to Full Stack Web Development and UI/UX(Job-roles and skillset for full stack and UI/UX)	R-3	RW-10	Discuss about the Introduction, tools, applications, job roles, and skill set of Full stack web development	Students would be able to understand the tools, applications, job roles, and skill set of Full stack web development	Demonstration with PowerPoint presentation	website development
		SPILL OVER						
Week 14	Lecture 28				Spill Over			
Week 15	Lecture 29				Spill Over			
	Lecture 30				Spill Over			

Scheme for CA:

CA Category of this Course Code is:A0303 (3 out of 3)

Component	Weightage (%)	Mapped CO(s)
Certification - MOOCs 1	33.33	CO1, CO2
Certification - MOOCs 2	33.33	CO3, CO4
Portfolio	33.33	CO5, CO6

Details of Academic Task(s)

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

Academic Task	Objective	Detail of Academic Task	Nature of Academic Task (group/individuals)	Academic Task Mode	Marks	Allottment / submission Week
Certification - MOOCs 1	To assess the student based on their profile creation and performance in the MOOC registered during the CSE111 course.	Students need to produce the certificate earned for completing the MOOC that was assigned in the CSE111 course. Students will be evaluated on the basis of their performance. There will be 70% weightage given MOOC and 30% to profile creation of Unit 1 and Unit 2 syllabuses can be referred for profile creation.	Individual	Online	100	3 / 5
Certification - MOOCs 2	To make students aware about cohorts and the availability of online learning platforms, to prepare them for placements.	Students need to produce the certificate earned for completing the Cohort MOOC. Students will be evaluated based on their performance on Cohort MOOC completion during course duration. Unit 3 and Unit 4 syllabuses can be referred for profile creation.	Individual	Online	100	4 / 9
Portfolio	To prepare and assess Student for the creation of a dream CV, a video CV, and a future three-year road map	Students will be evaluated on the basis of their video presentations of their dream CVs. The video will be uploaded to YouTube in unlisted mode by the student. A student will also create a Gantt chart of his or her career path plan. There will be 70% weightage given to CV generation and three year career roadmap and 30% for the profile creation. Unit 5 and Unit 6 syllabus can be referred for profile creation.	Individual	Online	100	8 / 12